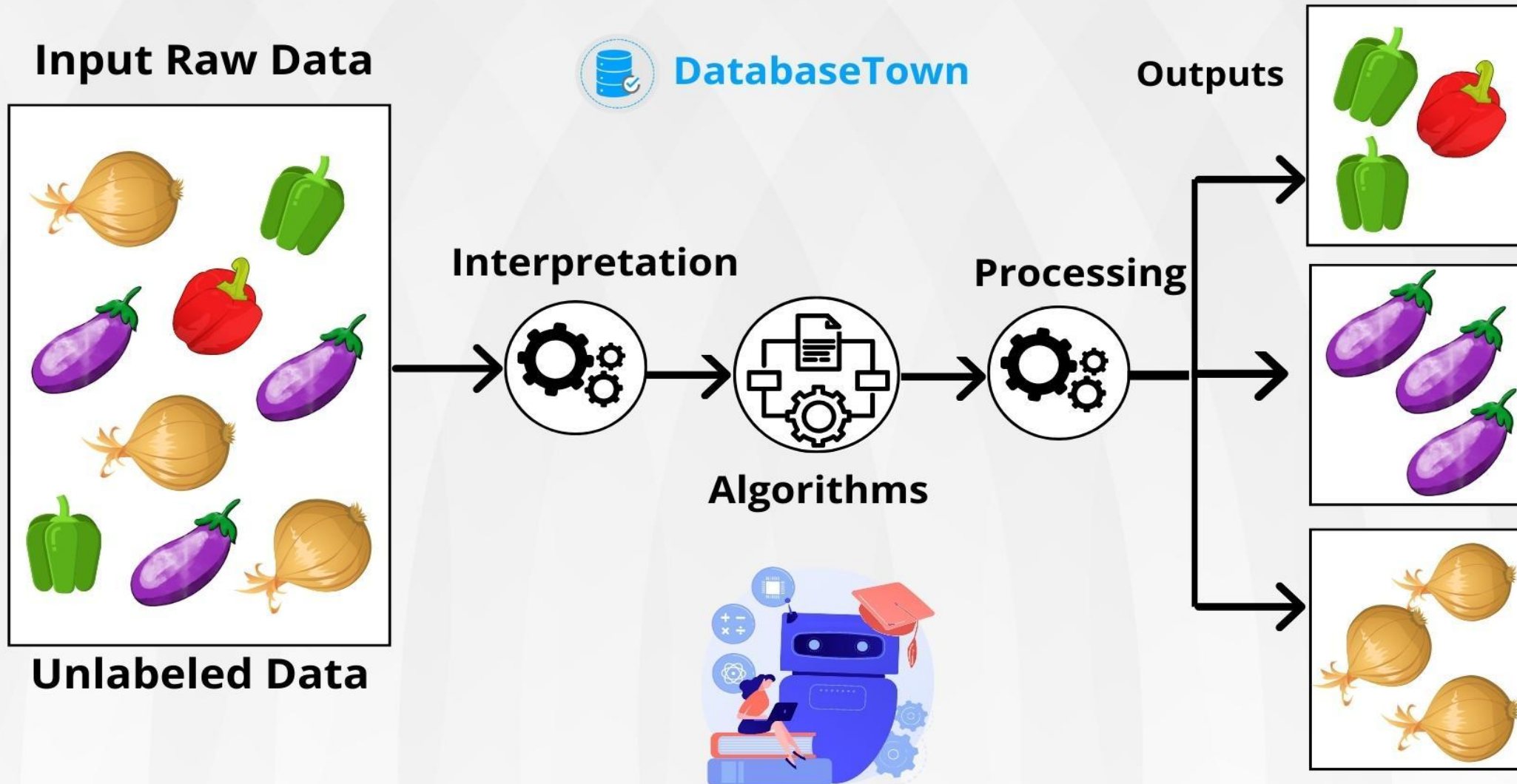


UNSUPERVISED LEARNING

Unsupervised learning is a type of machine learning where the algorithm learns from unlabeled data without any predefined outputs or target variables.

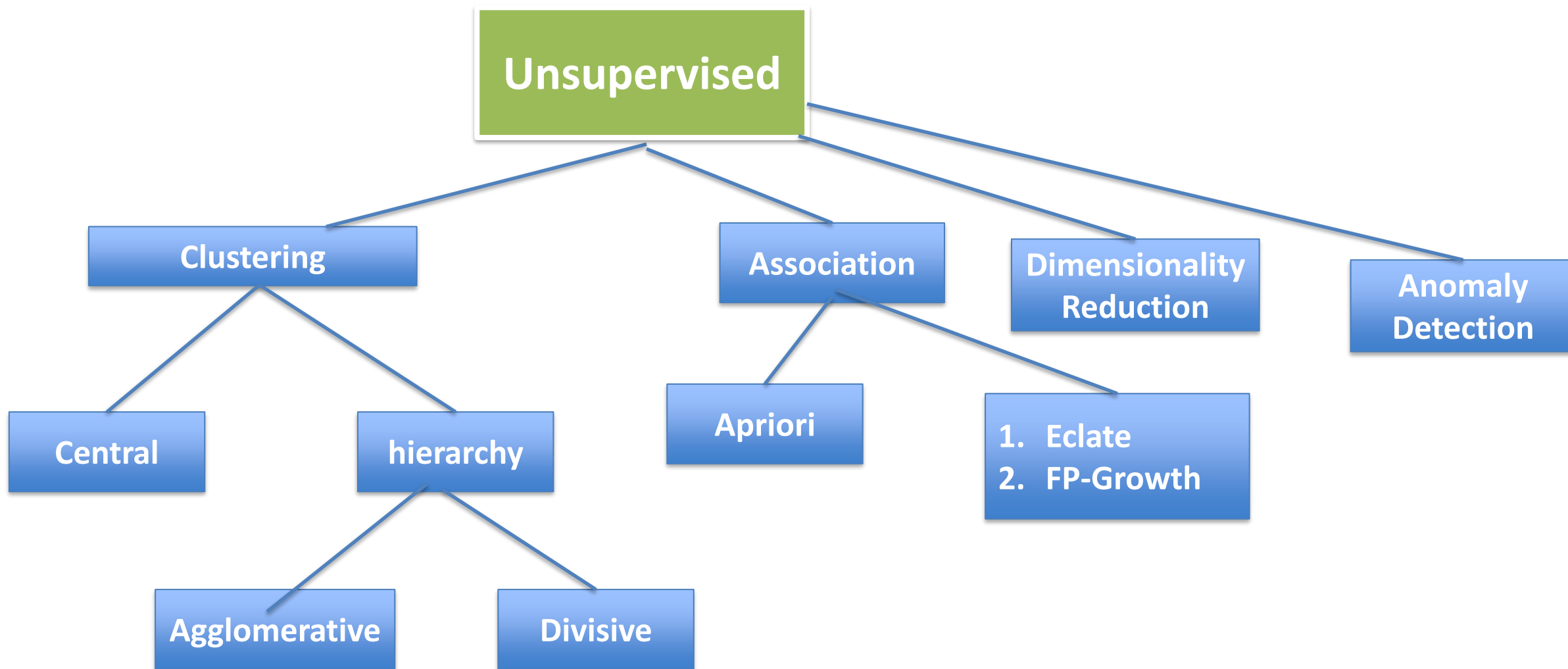


Supervised vs Unsupervised Learning :

	Supervised Learning	Unsupervised learning
Objective	To approximate a function that maps inputs to outputs based out example input-output pairs.	To build a concise representation of the data and generate imaginative content from it.
Accuracy	Highly accurate and reliable.	Less accurate and reliable.
Complexity	Simpler method.	Computationally complex.
Classes	Number of classes is known.	Number of classes is unknown.
Output	A desired output value (also called the supervisory signal).	No corresponding output values.

Introduction to Unsupervised Learning

- Unsupervised learning is a type of machine learning where the algorithm is given data without explicit labels or targets.
- The goal is to discover patterns, structures, or relationships in the data.
- Unlike supervised learning, there is no 'correct' output to compare with.



Types of Unsupervised Learning

1. Clustering:

Grouping data points into clusters based on similarity (e.g., K-Means, Hierarchical Clustering).

2. Association:

Finding relationships between variables in large datasets (e.g., Apriori, Eclat).

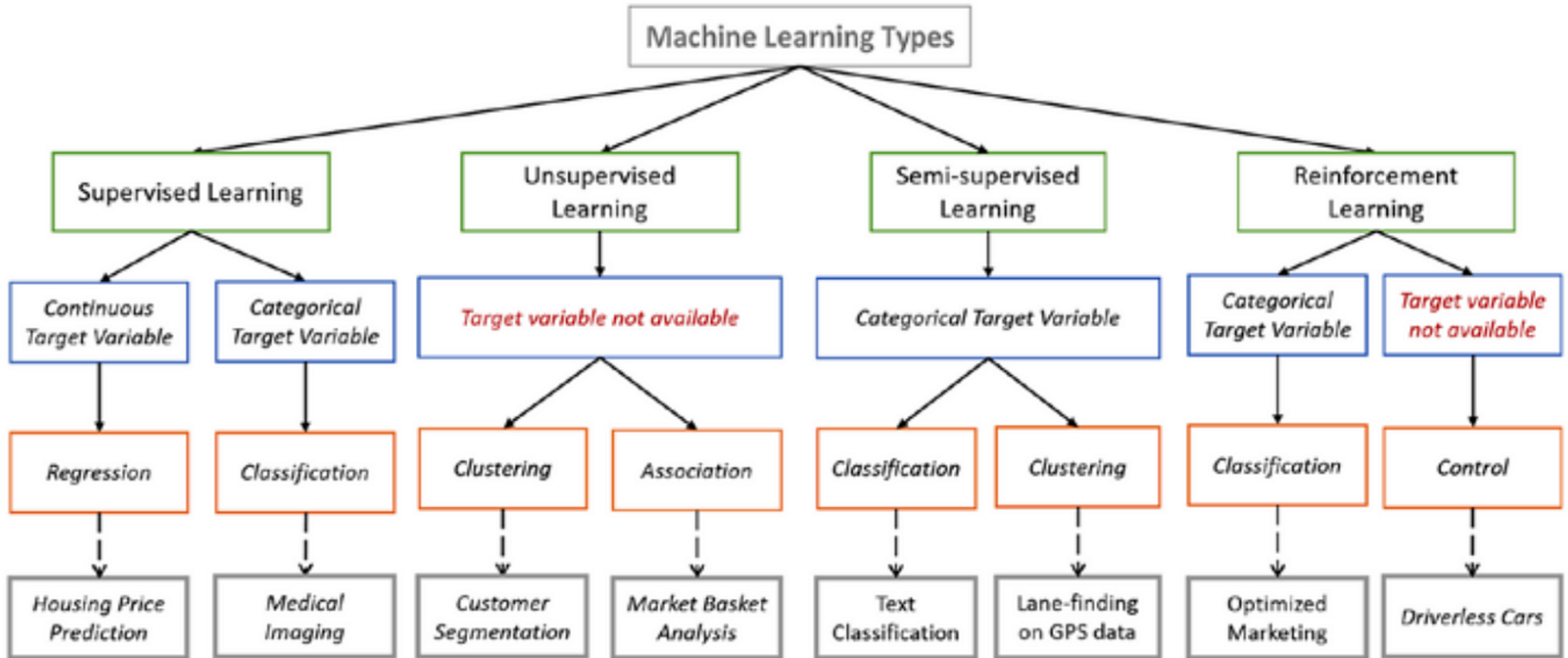
3. Dimensionality Reduction:

Reducing the number of variables while retaining key information (e.g., PCA, t-SNE).

Real-World Examples

- **1. Clustering:** Customer segmentation in marketing to target specific groups.
- **2. Association:** Market basket analysis to discover itemset frequently purchased together.
- **3. Dimensionality Reduction:** Image compression and data visualization.

Types of Unsupervised Learning (Diagram)



Applications of Unsupervised Learning

Unsupervised learning has many applications in:

1. Anomaly Detection: Identifying outliers or unusual data points.
2. Recommendation Systems: Suggesting products or services based on customer preferences.
3. Fraud Detection: Discovering fraudulent activity through patterns in transaction data.